

# HealthySnap

## Requirements & Project Report

### 1. Project Overview

| Field        | Details                                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Name | HealthySnap                                                                                                                                           |
| Type         | Mobile Application (Graduation Thesis)                                                                                                                |
| Platform     | iOS & Android (React Native)                                                                                                                          |
| Description  | A smart nutrition tracking and personal health management app with AI-powered food recognition, personalized health coaching, and community features. |
| Deadline     | August 2026                                                                                                                                           |

**HealthySnap** is a personal nutrition tracking app that makes healthy eating simple and smart. Users can scan food photos or barcodes to automatically retrieve nutritional information, or manually log meals with full details including calories, protein, carbs, and fat. The app personalizes the experience by calculating BMI and daily calorie targets based on the user's body stats, health goals, and activity level. Users with health conditions such as diabetes or hypertension will receive appropriate food warnings tailored to their needs. Beyond meal tracking, **HealthySnap** helps users plan weekly menus, track water intake and exercise, and view nutrition trends through weekly and monthly reports. A community feature allows users to share meals and follow others to stay motivated on their health journey.

### 2. Tech Stack

| Layer           | Technology                       |
|-----------------|----------------------------------|
| Mobile App      | React Native + Expo + TypeScript |
| Navigation      | Expo Router                      |
| Backend         | Node.js + Express                |
| Database        | MongoDB Atlas                    |
| Authentication  | JWT (JSON Web Token)             |
| AI Scan (Photo) | Claude API + Nutritionix API     |
| Barcode Scan    | Open Food Facts API              |
| Avatar Storage  | Cloudinary                       |

|                   |                    |
|-------------------|--------------------|
| Email Service     | Nodemailer + Gmail |
| API Documentation | Swagger UI         |

### 3. System Architecture

Frontend (React Native) communicates with Backend (Node.js) via REST API.

Backend handles all business logic, validation, and calculations.

Frontend handles UI rendering and UX validation only.

All calculations (BMI, TDEE, calorie totals) are performed on the backend.

### 4. Requirements List

#### 4.1 Authentication

| # | Feature                                   | Status |
|---|-------------------------------------------|--------|
| 1 | User Registration (name, email, password) | ✓ Done |
| 2 | User Login with JWT token                 | ✓ Done |
| 3 | Forgot Password via OTP email             | ✓ Done |
| 4 | Logout                                    | ✓ Done |

#### 4.2 User Profile

| #  | Feature                                                   | Status |
|----|-----------------------------------------------------------|--------|
| 5  | View profile with BMI and TDEE stats                      | ✓ Done |
| 6  | Update health info (gender, age, weight, height)          | ✓ Done |
| 7  | Set health goal (lose weight / gain muscle / eat healthy) | ✓ Done |
| 8  | Set activity level (sedentary / moderate / active)        | ✓ Done |
| 9  | Declare health conditions (diabetes, hypertension)        | ✓ Done |
| 10 | Auto-calculate BMI from weight + height                   | ✓ Done |
| 11 | Auto-calculate TDEE and calorie goal                      | ✓ Done |

|    |                                                       |                                                                                                      |
|----|-------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 12 | Upload avatar via Cloudinary                          |  In Progress      |
| 13 | Change name                                           |  Done (API ready) |
| 14 | Change password with OTP verification                 |  Done             |
| 15 | Height/weight picker with unit toggle (cm/ft, kg/lbs) |  In Progress      |

### 4.3 Meal Tracking

| #  | Feature                                                 | Status                                                                                     |
|----|---------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 16 | Add meal manually (name, calories, protein, carbs, fat) |  Done   |
| 17 | Select meal type (breakfast, lunch, dinner, snack)      |  Done   |
| 18 | View today's meals by meal type                         |  Done   |
| 19 | View meal history                                       |  Done   |
| 20 | Delete meal                                             |  Done   |
| 21 | Daily calorie + macro totals calculated by backend      |  Done |
| 22 | View meals from previous dates                          |  Done |

### 4.4 Home Dashboard

| #  | Feature                                           | Status                                                                                     |
|----|---------------------------------------------------|--------------------------------------------------------------------------------------------|
| 23 | Daily calorie progress bar                        |  Done |
| 24 | Macro breakdown (carbs, fat, protein)             |  Done |
| 25 | 7-day date selector (clickable)                   |  Done |
| 26 | Streak counter (consecutive logged days)          |  Done |
| 27 | Meal diary grouped by meal type                   |  Done |
| 28 | Smart meal suggestion based on remaining calories |  Done |

### 4.5 Scan Feature

| #  | Feature                                                 | Status                                                                                        |
|----|---------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 29 | Barcode/QR scan for packaged food (Open Food Facts API) |  Pending |

|    |                                                  |                                                                                             |
|----|--------------------------------------------------|---------------------------------------------------------------------------------------------|
| 30 | Photo scan with AI food recognition (Claude API) |  Pending |
| 31 | Auto-fill nutrition info from scan result        |  Pending |

## 4.6 AI Coach

| #  | Feature                                              | Status                                                                                      |
|----|------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 32 | Analyze weekly eating patterns                       |  Pending |
| 33 | Detect patterns (skipping meals, low protein, etc.)  |  Pending |
| 34 | Personalized suggestions based on history            |  Pending |
| 35 | Warn about foods incompatible with health conditions |  Pending |

## 4.7 Meal Planning

| #  | Feature                                 | Status                                                                                        |
|----|-----------------------------------------|-----------------------------------------------------------------------------------------------|
| 36 | Create weekly meal plan                 |  Pending |
| 37 | Calculate total nutrition for meal plan |  Pending |
| 38 | Meal reminders by meal type             |  Pending |

## 4.8 Health Companion

| #  | Feature                     | Status                                                                                        |
|----|-----------------------------|-----------------------------------------------------------------------------------------------|
| 39 | Water intake tracking       |  Pending |
| 40 | Exercise activity logging   |  Pending |
| 41 | Daily water/exercise totals |  Pending |

## 4.9 Statistics & Reports

| #  | Feature                        | Status                                                                                        |
|----|--------------------------------|-----------------------------------------------------------------------------------------------|
| 42 | Nutrition trends by week/month |  Pending |

|    |                           |                                                                                             |
|----|---------------------------|---------------------------------------------------------------------------------------------|
| 43 | Charts and visualizations |  Pending |
|----|---------------------------|---------------------------------------------------------------------------------------------|

## 4.10 Community

| #  | Feature             | Status                                                                                      |
|----|---------------------|---------------------------------------------------------------------------------------------|
| 44 | Share meals to feed |  Pending |
| 45 | Follow other users  |  Pending |

## 4.11 Localization

| #  | Feature                     | Status                                                                                                       |
|----|-----------------------------|--------------------------------------------------------------------------------------------------------------|
| 46 | English language (primary)  |  Done                     |
| 47 | Vietnamese language support |  Pending (if time allows) |

## 5. API Endpoints

| Method | Endpoint                   | Description              |
|--------|----------------------------|--------------------------|
| POST   | /api/auth/register         | Register new user        |
| POST   | /api/auth/login            | Login user               |
| GET    | /api/auth/me               | Get current user         |
| POST   | /api/meals                 | Add meal                 |
| GET    | /api/meals?date=YYYY-MM-DD | Get meals by date        |
| GET    | /api/meals/history         | Get meal history         |
| DELETE | /api/meals/:id             | Delete meal              |
| GET    | /api/profile               | Get profile + BMI + TDEE |
| PUT    | /api/profile               | Update profile           |
| POST   | /api/user/avatar           | Upload avatar            |
| PUT    | /api/user/name             | Change name              |
| POST   | /api/user/send-otp         | Send OTP to email        |

|      |                          |                         |
|------|--------------------------|-------------------------|
| POST | /api/user/reset-password | Reset password with OTP |
|------|--------------------------|-------------------------|

## 6. Project Progress Summary

| Metric           | Value       |
|------------------|-------------|
| Total Features   | 47          |
| Completed        | 22 features |
| In Progress      | 3 features  |
| Pending          | 22 features |
| Overall Progress | ~47%        |

## 7. Validation Rules

| Field    | Rule                                                       |
|----------|------------------------------------------------------------|
| Name     | Min 2 characters, letters and spaces only                  |
| Email    | Valid format (example@domain.com), stored as lowercase     |
| Password | Min 6 characters, at least 1 uppercase letter and 1 number |
| Age      | Between 10 and 120                                         |
| Weight   | Between 20 and 300 kg                                      |
| Height   | Between 50 and 250 cm                                      |